

Report: WhatsApp API Usage, Pricing & Cost Optimization

1. Objective

To evaluate WhatsApp messaging options for **OTP verification, event invitations, reminders, and broadcasts**, understand **pricing differences (India vs UAE)**, clarify the **24-hour conversation window**, and assess **realistic costs & savings** using the **official WhatsApp Business API**.

2. Core Requirements

- WhatsApp OTP verification
 - Event invitations & reminders
 - Broadcast messaging
 - Official WhatsApp Business API (no risk of bans)
 - Cost optimization if possible
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3. WhatsApp API Providers Evaluated

Top 3 Suitable Options

1. **Twilio WhatsApp API**
 - Best for custom workflows and developer control
 - No monthly platform fee
2. **WATI**

- Easy dashboard, fast setup
- Monthly platform fee

3. Gupshup

- Cost-effective, good API + automation balance
- Monthly platform fee

✓ All three use the **official WhatsApp Business API** and meet functional requirements.

4. Key Pricing Reality (Most Important Point)

! WhatsApp pricing depends on the destination country of the user, not:

- Provider country
- Company location
- Server location

➡ Messages sent to **UAE numbers** are always charged at **UAE rates**, even if using an **Indian provider**.

5. Why India Is Cheap & UAE Is Expensive

- **India:** Massive WhatsApp usage, low telecom cost, high volume → very low per-message price
 - **UAE:** Smaller market, higher telecom & regulatory costs → higher per-message price
 - Currency difference magnifies the gap (1 AED ≈ ₹22–23)
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6. Official WhatsApp API Pricing (Approx)

UAE (Meta Pricing)

- OTP / verification template: **~AED 0.20**
 - Broadcast / reminder template: **~AED 0.25**
 - Session message (within 24h): **~AED 0.07–0.10**
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7. Usage Assumptions Used in Analysis

- OTPs: **100/day → 3,000/month**
 - Broadcast messages: **20,000/month**
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8. Monthly Cost – UAE (Official WhatsApp API)

Message Cost (Cannot Be Reduced)

- OTP: $3,000 \times 0.20 = \text{AED } 600$
- Broadcast: $20,000 \times 0.25 = \text{AED } 5,000$

➡ **Minimum WhatsApp message cost = ~AED 5,600/month**

9. Using Indian Providers for UAE Numbers

Indian providers can **reduce platform/agency fees**, but **cannot reduce message cost**.

Provider	Platform Fee	Total Monthly Cost (Approx)
Twilio (India)	AED 0	~AED 5,600

Gupshup (India) ~AED 300 ~AED 5,900

WATI (India) ~AED 400–500 ~AED 6,000

➡ Message cost remains **UAE Meta pricing**.

10. Clarification: 24-Hour Conversation Window

How It Works

1. **Outbound template message** (OTP / invite) → **Always paid**
2. If user replies → **24-hour session window opens**
3. Messages sent within 24 hours:
 - No template required
 - Charged as **session messages (cheaper)**
4. After 24 hours without reply:
 - Session closes
 - Next message must be a **paid template**

Important Truth

- The 24-hour window **does not stay open until event day**
 - OTP reply alone **does not guarantee free future messages**
 - Every new initiation after 24 hours = **paid**
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11. Scenario Analysis (40% Users Reply)

Assumptions

- Total users: 20,000
- 40% reply = 8,000 users
- 60% don't reply = 12,000 users

Total Cost (Approx – UAE)

- OTP messages
- Event info (mixed session + template)
- Reminder before event (template for all)
- Final follow-up (session for those who replied)

➡ **Total ≈ AED 13,280**

Insight

- 24-hour window helps reduce **follow-up costs**
- Overall savings are **~10–15%**, not dramatic
- Broadcasts and reminders still dominate cost

12. Why AED 2,000/Month Is Unrealistic

For official WhatsApp API:

- OTP + 20,000 broadcasts alone exceed AED 5,600
- AED 2,000 is possible only if:
 - Very low volume
 - Heavy use of email/SMS instead

- OR non-official WhatsApp tools (high risk)
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13. Realistic Cost-Reduction Options

✓ Allowed & Safe

- Use WhatsApp only for **OTP + critical reminders**
- Shift marketing & non-critical communication to **email**
- Combine multiple updates into **single messages**
- Encourage replies to leverage session messages

✗ Not Allowed / High Risk

- Routing UAE traffic as India
 - Unofficial bulk WhatsApp tools
 - Fake sender numbers
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14. Final Conclusion

- WhatsApp API pricing is **controlled by Meta and country-based**
- UAE messaging is inherently expensive compared to India
- Indian providers can reduce **platform fees**, not **message cost**
- 24-hour window helps but **does not eliminate charges**
- Minimum realistic cost for current usage is **~AED 5,600/month**
- Significant cost reduction requires **reducing WhatsApp usage**, not changing providers

